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ON

Federated Learning Framework for IoT Intrusion Detection

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CONTENTS

S.No.	Title	Page No.
1	Abstract	5
2	List of Figures and Tables	6
3	Introduction	7-9
4	Methodology	10-12
5	Implementation	13-15
6	Major Findings / Outcomes / Results	16-17
7	Conclusion and Future Scope	18
	References	19-20

DECLARATION

We hereby certify that the work which is being presented in the project report entitled "Federated Learning Framework for IoT Intrusion Detection" in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering (Computer Science and Engineering), submitted in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of Dr. Monika Aggarwal. The matter presented in this project report has not been submitted to any other university/institute for the award of any degree.

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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ABSTRACT

As the Internet of Things (IoT) continues to grow at an unprecedented scale, traditional centralized Intrusion Detection Systems (IDS) are increasingly struggling with fundamental challenges, including data privacy concerns, high communication latency, bandwidth limitations, and the inherent risk of a single point of failure. To address these critical issues, Federated Learning (FL) has emerged as a powerful and principled decentralized machine learning paradigm.

This project report explores the development and evaluation of a Federated Learning-based framework for IoT security and intrusion detection. It examines the three primary categories of Federated Learning — Horizontal Federated Learning (HFL), Vertical Federated Learning (VFL), and Federated Transfer Learning (FTL) — to assess their applicability and effectiveness across diverse IoT network configurations.

The study specifically investigates FL-based IDS deployments across two critical and contrasting real-world environments: Smart Cities and Healthcare (Internet of Medical Things — IoMT). In Smart City deployments, the framework is designed to manage large-scale, heterogeneous networks encompassing traffic control systems, smart energy grids, and environmental sensors, where the primary objectives are fast processing, low latency, and efficient bandwidth utilization. In the Healthcare domain, the framework emphasizes compliance with stringent regulatory frameworks (such as HIPAA and GDPR) and the robust protection of highly sensitive patient health information generated by wearable devices and clinical monitoring systems.

A comprehensive comparative analysis reveals that Smart City implementations prioritize scalability and rapid response times, while Healthcare applications necessitate more sophisticated and aggressive privacy-preserving mechanisms. The results

demonstrate that decentralized model training combined with secure global aggregation can achieve detection performance comparable to centralized approaches, while substantially reducing communication overhead and eliminating the risks associated with raw data transmission.

Keywords: Federated Learning, Internet of Things (IoT), Intrusion Detection System (IDS), Smart City, Healthcare, Data Privacy, Anomaly Detection, Machine Learning, GDPR, HIPAA.

LIST OF FIGURES AND TABLES

List of Figures

Figure No.	Title
Figure 1	Vertical Federated Learning (VFL) Architecture
Figure 2	Horizontal Federated Learning (HFL) Architecture
Figure 3	Transfer Federated Learning (TFL) Architecture
Figure 4	Smart City vs. Healthcare IoT — Comparative Overview Table

List of Tables

Table No.	Title
Table 1	Smart City vs. Healthcare IoT Deployment Comparison

1. INTRODUCTION

1.1 Background

The Internet of Things (IoT) represents a paradigm-defining evolution in global digital infrastructure. At its core, IoT is a vast network of physical objects — encompassing sensors, meters, vehicles, wearable devices, industrial machinery, and smart home appliances — all embedded with electronics, software, and communication capabilities that enable them to collect, transmit, and exchange data over the internet. These interconnected devices are deployed across diverse sectors including smart cities, healthcare systems (Internet of Medical Things — IoMT), and industrial control networks (Industrial IoT — IIoT).

Unlike traditional computing nodes, IoT devices are inherently resource-constrained, characterized by limited processing power, restricted memory, minimal battery capacity, and intermittent network connectivity. This architectural heterogeneity, while enabling the deployment of billions of devices at the edge of the network, simultaneously introduces a broad and complex attack surface. IoT networks are increasingly targeted by sophisticated cyber threats including botnets, Distributed Denial of Service (DDoS) attacks, man-in-the-middle (MITM) attacks, and unauthorized data exfiltration attempts.

1.2 Problem Statement

Centralized Intrusion Detection Systems (IDS) — the traditional security paradigm — are fundamentally ill-suited to protect modern IoT ecosystems. Centralizing security analysis requires the transmission of massive volumes of raw network traffic data to a central server, which introduces several critical challenges:

- Data Privacy Risks: Transmitting sensitive device data to a central repository exposes private information to potential breaches and violates data protection regulations such as GDPR and HIPAA.

- High Communication Latency and Bandwidth Overhead: Routing all IoT traffic through a central server creates significant network bottlenecks, introducing unacceptable delays in threat detection.

- Single Point of Failure: Centralizing security analysis creates a systemic vulnerability; if the central server is compromised or becomes unavailable, the entire security infrastructure collapses.
- Scalability Constraints: The exponential growth in the number of deployed IoT devices makes centralized processing increasingly computationally prohibitive.

1.3 Proposed Solution: Federated Learning for IoT IDS

Federated Learning (FL), introduced by McMahan et al. (2017), represents a fundamentally different approach to distributed machine learning. Rather than transmitting raw data to a central server, FL enables individual devices — referred to as federated clients — to train machine learning models locally using their own datasets. Only the resulting model updates (parameters or gradients), not the raw data, are transmitted to a central aggregation server. This data-to-model approach, as opposed to the traditional model-to-data approach, inherently preserves data privacy by ensuring that sensitive information never leaves its device of origin.

The FL framework can be conceptualized as a global neighborhood watch system for IoT security, where every smart device learns to recognize and flag suspicious activity independently, without sharing its private data streams with any external entity. Instead, it contributes distilled "knowledge updates" — mathematical representations of learned threat patterns — to a central aggregator, which synthesizes these contributions into a continually improving global intrusion detection model.

1.4 Objectives

- To propose and evaluate a Federated Learning-based Intrusion Detection System (FL-IDS) specifically designed for heterogeneous IoT environments.
- To analyze and compare the three primary categories of Federated Learning — Horizontal, Vertical, and Transfer — in terms of their suitability for different IoT network topologies.
- To demonstrate the practical applicability of the FL-IDS framework through two real-world use case scenarios: Smart City IoT infrastructure and Healthcare IoT (IoMT) networks.
- To conduct a systematic comparative analysis between Smart City and Healthcare FL-IDS deployments, identifying the key differences in data sensitivity, threat profiles, and operational requirements.
- To identify the limitations of current approaches and outline directions for future research.

1.5 Scope of the Report

This report covers the theoretical foundations of Federated Learning and Intrusion Detection Systems, the taxonomy of FL approaches, FL-based IDS architecture, and two detailed use case analyses. It is grounded in a comprehensive review of recent literature (2021–2026) and draws upon the research paper "Federated Learning Framework for IoT Intrusion Detection" authored by Rhythm and Prisha Aggarwal, Chitkara University.

2. METHODOLOGY

2.1 Research Approach

This project adopts a qualitative and analytical research methodology. The work is grounded in a systematic review of existing literature on Federated Learning, IoT security, and Intrusion Detection Systems. The methodology can be categorized into the following phases:

2.2 Literature Review

A comprehensive review of recent peer-reviewed publications, conference papers, and technical reports (spanning 2017 to 2026) was conducted to establish the theoretical foundation of this work. Key contributions reviewed include:

- McMahan et al. (2017): The seminal paper introducing the Federated Averaging (FedAvg) algorithm, which forms the algorithmic backbone of most FL systems.
- Truong et al. (2021): An analysis of privacy preservation in FL from the GDPR perspective, which informed the privacy analysis in the use case scenarios.
- Rehman et al. (2026): Introduced Incremental Federated Learning using LSTM models for evolving IoT threat landscapes, tested on the CICIoMT2024 dataset.
- Fed-DTCN (2026): An unsupervised FL framework using dual-encoder contrastive learning for zero-day attack detection in IoT networks.
- Fares et al. (2025): Demonstrated that FL with Tab Transformer architectures achieves state-of-the-art IDS accuracy on high-dimensional IoT traffic data.
- Mothukuri et al. (2024/2025): Improved GRU-based DDoS detection and proposed a federated SVM model that matches centralized benchmarks on low-power edge devices.

2.3 Classification of Federated Learning

The three main categories of Federated Learning were studied and classified based on data distribution characteristics, use cases, and privacy mechanisms:

A. Horizontal Federated Learning (HFL): Applied when multiple clients share the same feature space but possess different data instances. This is the most common FL configuration, applicable to scenarios where organizations serve the same population segment (e.g., multiple hospitals with similar patient record structures). In IoT contexts, HFL is used when edge devices generate data with identical feature sets — for instance, network traffic logs from multiple city zones sharing the same metadata format.

B. Vertical Federated Learning (VFL): Applied when clients share the same sample space but hold different feature sets. VFL is suited for cross-organizational collaborations where different entities hold complementary information about the same set of users or events — for example, a healthcare provider with clinical data and an insurance company with financial records collaborating to build a joint predictive model without disclosing their respective datasets.

C. Federated Transfer Learning (FTL): Applied when clients differ in both sample and feature space, making direct model sharing infeasible. FTL leverages transfer learning techniques to enable knowledge transfer across parties with heterogeneous data structures, such as when a new IoT deployment lacks sufficient labeled data and must leverage a pre-trained model from a different but related domain.

2.4 Use Case Analysis Framework

Two distinct real-world deployment environments — Smart City IoT and Healthcare IoT — were selected to validate the practical applicability of the proposed FL-IDS framework. For each environment, the following analytical dimensions were examined:

- Network topology and device heterogeneity
- Data sensitivity classification and applicable regulatory frameworks
- Characteristic threat profiles and attack vectors
- Federated client architecture and aggregation strategy
- Privacy-preserving mechanism requirements
- Communication frequency and model update scheduling

2.5 Comparative Analysis

A structured comparative analysis was conducted between the Smart City and Healthcare IoT deployments across multiple parameters including data sensitivity, primary threat types, federated client definition, model update frequency, and regulatory compliance requirements. This analysis is presented in Chapter 5 and summarized in a comparative table (Figure 4 / Table 2) to provide a clear, side-by-side evaluation of the two environments.

3. IMPLEMENTATION

3.1 FL-IDS Framework Architecture

The proposed Federated Learning-based Intrusion Detection System follows a three-tier architecture consisting of IoT edge devices (federated clients), local edge gateways, and a central federated aggregation server.

Tier 1 — IoT Edge Devices: Individual IoT devices (sensors, cameras, wearables, etc.) continuously monitor local network traffic and generate network flow records. These devices perform lightweight feature extraction from raw traffic and serve as the primary data sources for local model training.

Tier 2 — Local Edge Gateways: Each zone or department is served by an edge gateway that aggregates data from its connected IoT devices, performs local model training on the collected network flow data using the current global model as the starting point, and periodically transmits model updates (not raw data) to the central aggregation server.

Tier 3 — Central Aggregation Server: The central server receives model updates from all participating edge gateways, applies the Federated Averaging (FedAvg) algorithm to produce an updated global model, and distributes the improved global model back to all participating clients. This cycle repeats continuously, enabling the global model to improve with each aggregation round.

3.3 Smart City IoT Deployment

In the Smart City scenario, the FL-IDS is deployed across distinct city zones, where each zone's edge gateway functions as a federated client. Local intrusion detection models are trained on the network traffic of that zone's IoT infrastructure — traffic signals, smart meters, environmental sensors, and surveillance cameras — without transmitting any raw traffic data outside the local network perimeter.

The Smart City deployment is optimized for:

- High-frequency model updates: Due to the dynamic, high-volume nature of smart city network traffic, model updates are transmitted to the central aggregation server at frequent intervals (e.g., hourly) to ensure rapid adaptation to emerging threats.
- Scalability: The framework is designed to accommodate thousands of federated clients across a metropolitan area, with hierarchical aggregation strategies to manage communication overhead.
- DDoS and Botnet Detection: Local models are specifically trained to identify volumetric attack signatures and anomalous traffic patterns consistent with botnet activity or MITM attacks at the zone level.

3.4 Healthcare IoT Deployment

In the Healthcare IoT scenario, each hospital or clinic functions as a separate federated client. Local intrusion detection models are trained exclusively on the network traffic generated within that institution's internal IoMT infrastructure — electronic health record (EHR) systems, infusion pumps, wearable biosensors, and diagnostic imaging

equipment — without any patient health information (PHI) leaving the institution's secure network boundary.

The Healthcare deployment is optimized for:

- Aggressive privacy preservation: Differential privacy mechanisms are applied to model updates before transmission to the federated aggregation server, ensuring that no information about individual patient data can be inferred from the shared updates.
- Scheduled aggregation: Model updates are scheduled during low-activity clinical periods to minimize disruption to critical healthcare operations.
- Ransomware and Unauthorized Access Detection: Local models are trained to identify anomalous patterns consistent with ransomware propagation, unauthorized device access attempts, and data exfiltration behaviors.

4.5 Types of Federated Learning — Comparative Table

Parameter	Horizontal FL	Vertical FL	Transfer FL
Data Distribution	Same features, diff. samples	Diff. features, same samples	Diff. features & samples
Use Case	Multiple orgs, same data type	Cross-org complementary data	Knowledge transfer across domains
IoT Suitability	High (most common)	Moderate	For new/data-sparse deployments
Privacy Complexity	Moderate	High	High

Table 1: Comparison of FL Types

4. MAJOR FINDINGS / OUTCOMES / RESULTS

4.1 Key Findings from Literature Review

The comprehensive literature review conducted for this study yielded several significant findings regarding the state of the art in Federated Learning-based IoT intrusion detection:

- Federated Learning significantly outperforms traditional centralized IDS approaches in privacy preservation: By design, FL eliminates the need to transmit raw IoT traffic data to any centralized repository, making it fundamentally more privacy-preserving than any centralized approach.
- FL-based IDS achieves competitive detection accuracy: Fares et al. (2025) demonstrated that FL frameworks utilizing Tab Transformer architectures achieve state-of-the-art detection accuracy on benchmark IoT datasets, while FL-based IDS trained for 6G-enabled smart cities achieved up to 98% test accuracy.
- Communication overhead is substantially reduced: FL-based IDS systems trained for next-generation smart city deployments achieved 35% less communication overhead compared to traditional centralized FL systems, demonstrating the efficiency gains achievable through distributed model training.
- Incremental learning addresses evolving threat landscapes: Rehman et al. (2026) demonstrated that LSTM-based incremental FL allows the IDS to continuously adapt to new attack types while retaining knowledge of previously encountered threats — a critical capability for IoT environments where attack patterns evolve rapidly.
- Unsupervised FL can detect zero-day attacks: The Fed-DTCN framework (2026) demonstrated that contrastive learning-based FL approaches can detect previously unseen zero-day attacks, significantly outperforming supervised approaches in scenarios where labeled attack traffic is scarce.

4.2 Comparative Analysis: Smart City vs. Healthcare IoT

The following table presents a structured comparison of the two use case deployments across key operational parameters:

Parameter	Smart City IoT	Healthcare IoT
Device Types	Traffic sensors, smart meters, cameras	Wearables, infusion pumps, EHR terminals
Data Sensitivity	Moderate	Very High (PHI)
Primary Threats	DDoS, Botnet, MITM attacks	Ransomware, unauthorized access, data exfiltration
Federated Clients	City zones / districts	Hospitals / clinics
Update Frequency	High (frequent rounds)	Scheduled / lower frequency
Regulatory Framework	GDPR	GDPR, HIPAA
Privacy Mechanisms	Standard secure aggregation	Differential privacy + secure aggregation
FL Type Preferred	Horizontal FL	Horizontal FL + FTL for small clinics

Table 2: Smart City vs. Healthcare IoT Deployment Comparison

5. CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The rapid proliferation of IoT devices has fundamentally transformed the threat landscape of modern networks, rendering traditional centralized Intrusion Detection Systems increasingly inadequate in the face of distributed, privacy-sensitive, and resource-constrained environments. This project report has presented a comprehensive investigation of Federated Learning as a principled and practical alternative — one that enables IoT devices to collaboratively develop robust intrusion detection

The comparative analysis between the two scenarios further confirmed that federated intrusion detection is not a domain-specific solution but a broadly applicable security paradigm adaptable to diverse IoT environments with different data sensitivity profiles, threat landscapes, regulatory requirements, and operational constraints.

The study also highlighted the transformative potential of FL in enabling small, resource-constrained institutions — such as rural healthcare facilities or small city districts — to benefit from globally trained intrusion detection models that would otherwise be inaccessible to them, thereby democratizing IoT security at scale.

6.2 Future Scope

While this study establishes a robust foundation for FL-based IoT intrusion detection, several promising directions for future research have been identified:

- Blockchain-Enabled Model Update Auditing: The integration of blockchain technology to create immutable, tamper-proof audit trails of model updates would provide an additional layer of trust and accountability in federated aggregation processes, enabling verification of the integrity of model contributions from individual clients.
- Advanced Differential Privacy Techniques: Future work will explore adaptive differential privacy mechanisms that dynamically adjust the privacy budget based on the sensitivity of the data being processed, optimizing the privacy-utility tradeoff in real-time.

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